Meera Reddy

Thermal Systems Engineer

RESUME_1563

Location: Bhubaneswar, Odisha, India | Email: meera.reddy.resume_1563@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 4 years

PROFESSIONAL SUMMARY

Thermal Systems Engineer with 4 years of experience in thermal engineering, using Thermodynamics, Heat Transfer, ANSYS Fluent, CFD. Experienced in thermal design, energy efficiency and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Data Acquisition, CAD, CFD, ANSYS Fluent, Thermodynamics, Heat Transfer, Experimental Testing, HVAC, MATLAB, Heat Exchangers, Energy Audit

Methods: Thermal design, Energy efficiency, CFD validation

PROFESSIONAL EXPERIENCE

Thermal Systems Engineer | Urbanaut Engineering | Jan 2024 - Present

- Reduced simulated hotspot temperature by 39% through airflow and interface changes
- Reduced modeled process heat loss by 41% after insulation improvements
- Executed 545 controlled thermal test cases with uncertainty estimates
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate Thermal Systems Engineer | TerraNex Solutions | Jul 2022 - Dec 2023

- Executed 801 controlled thermal test cases with uncertainty estimates
- Improved heat-transfer performance by 36% in parametric design studies

PROJECTS

Compact Heat Exchanger Concept | CAD, HVAC, Heat Exchangers, ANSYS Fluent

- Compared thermal effectiveness and pressure-drop tradeoffs.
- Documented acceptance criteria and evaluated the design using energy efficiency and cfd validation.

Industrial Oven Energy Audit | Experimental Testing, MATLAB, Energy Audit, CFD

- Estimated losses, insulation upgrades and process energy balance.
- Documented acceptance criteria and evaluated the design using energy efficiency and cfd validation.

EDUCATION

B.Tech in Mechanical Engineering, Blue River College of Technology, 2022

Relevant Methods: Thermal design, Energy efficiency, CFD validation

CERTIFICATIONS

Applied Heat Exchanger Analysis (2024)